

Voyage brief - Antarctic ice-risk routing

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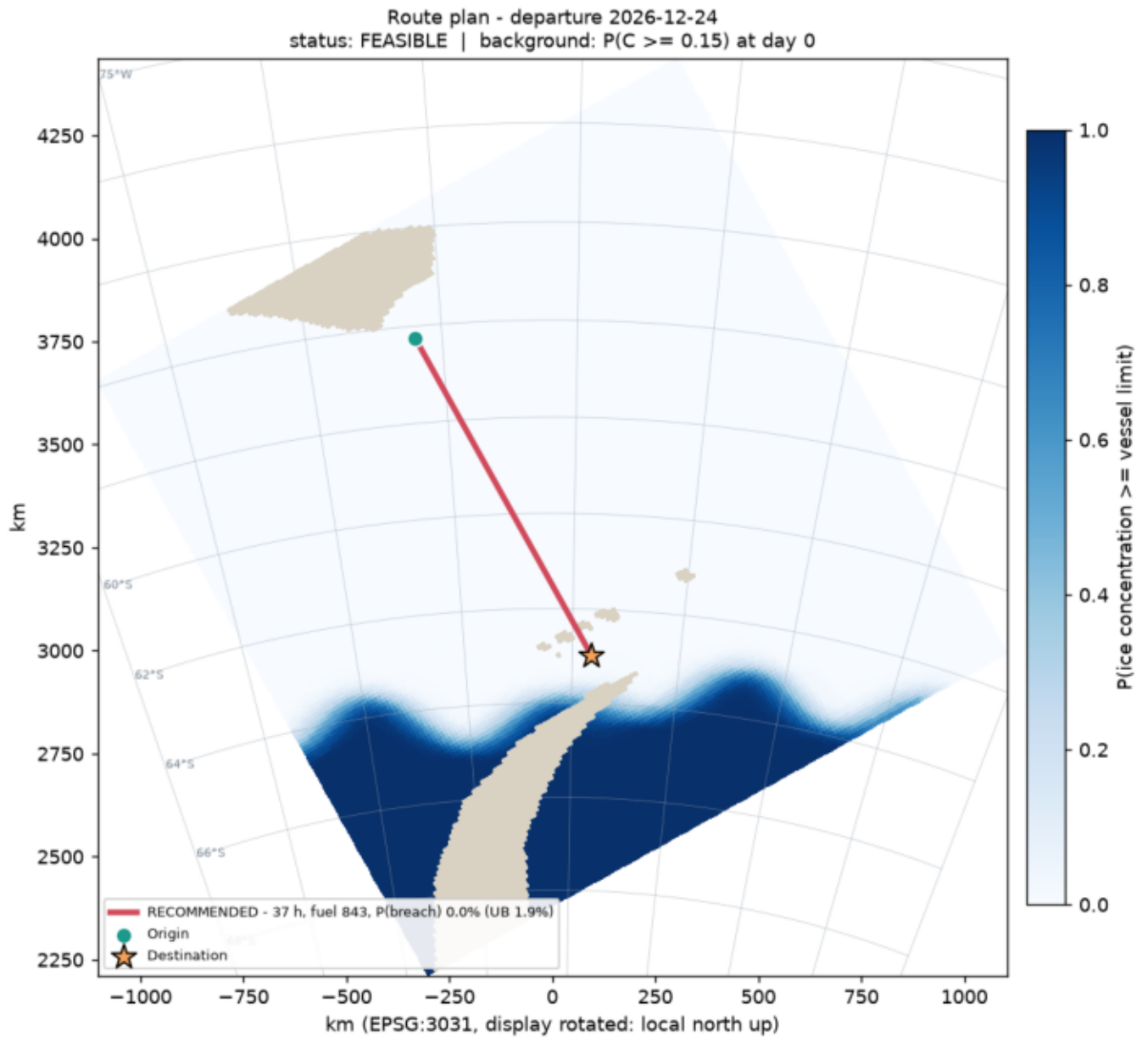
Route	(-56.3, -66.0) -> (-63.0, -59.0)
Departure (UTC)	2026-12-24T00:00:00
Decision	RECOMMENDED ROUTE
Risk budget	5.0% via wilson_upper at 95% confidence
Joint scenarios	200
Data mode	CONTROLLED_SYNTHETIC
Route labels	shortest_distance, minimum_time, minimum_fuel, risk_weighted_10, risk_weighted_100, risk_weighted_1000, scenario_17_optimal, scenario_130_optimal, scenario_154_optimal
P(breach)	0.0% (95% upper bound 1.9%)
Expected time	37.4 h (P10-P90 37.2-37.5 h)
Fuel index	843 (relative, not litres)
Distance	842 km
Beyond scenario horizon	0% of scenarios
Forecast trust horizon	>= 14 days (limited by evaluated leads)

Assumptions and limitations

- Vessel ice limit $\tau_v = 0.15$ and ice class 'REPLACE_WITH_VERIFIED_CLASS' are placeholders; derive them from verified vessel capability.
- Fuel is a relative index $d[1 + \lambda C^2]$; speed in ice $v(1 - kC)$ is an assumption (see sensitivity).
- Route risk is evaluated jointly across scenarios; it is a model estimate, not a guarantee.
- Tracked icebergs (USNIC) cover giant bergs only; smaller bergs and growlers are not represented.
- Geography in controlled-synthetic runs is schematic and must not be used for navigation.
- Recommended the minimum expected-fuel route among 1 of 1 candidates meeting the 5.0% risk budget (wilson_upper): P(breach)=0.0% (upper 1.9%), E[time]=37.4 h, E[fuel index]=843.

Research recommendation only - not a certified navigation route. Antarctic operations require authoritative ice information, vessel-specific operating limits, applicable ice-class guidance and qualified human oversight.

Route map



CONTROLLED_SYNTHETIC DATA - Research recommendation only - not a certified navigation route. Antarctic operations require authoritative ice information, vessel-specific operating limits, applicable ice-class guidance and qualified human oversight.

Candidate routes (same joint scenarios)

route	meets budget	km	E[h]	E[fuel]	P(breach)	95% UB
shortest_distance	yes	842	37.4	843	0.0%	1.9%